UNIT - V

THIN & THICK CYLINDERS, COLUMNS

Mechanics of Solids (23A03302) - handwritten class notes

Topics covered in this unit

- THIN CYLINDERS: seamless shells, longitudinal and circumferential stresses
- Hoop, longitudinal and volumetric strains; changes in diameter and volume
- Thin spherical shells; wire-wound thin cylinders
- THICK CYLINDERS: Lamé's equations, inside and outside pressures, compound cylinders
- COLUMNS: buckling and stability, Euler's formula for various end conditions
- Limitations of Euler's formula, Rankine's formula

Tip: thin cylinder: $\sigma_c = pd/2t$ and $\sigma_L = pd/4t$. Column: $P_{cr} = \pi^2 EI/L_e^2$.

PART A - THIN CYLINDERS AND SHELLS

1. Thin cylinders

A cylindrical shell is thin if the wall thickness t is small compared with the internal diameter d: $t < d/20$. Then the stress is assumed uniform across the thickness and the radial stress ($\approx p$ at the inside, 0 outside) is ignored.

Internal pressure p causes two principal stresses in the wall:

- Circumferential (hoop) stress σ_c - tries to split the cylinder along a length.

- Longitudinal stress σ_L - tries to tear the cylinder across a section.

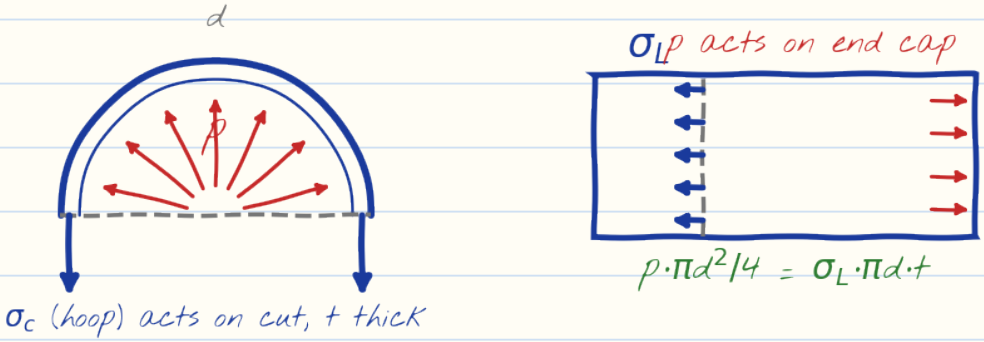


Fig: forces on a thin cylinder - hoop stress (left) and longitudinal stress (right)

2. Derivation of the stresses

Circumferential stress: cut the cylinder along a diameter, length L. Bursting force = p x d x L; resisting force = 2 σ_c t L.

$p\,d\,L = 2\,\sigma_c\,t\,L \Rightarrow \sigma_c = p\,d\,/\,2t$

Longitudinal stress: cut across. Bursting force = p x $\pi\,d^2/4$; resisting = σ_L x $\pi\,d\,t$.

$p\,\pi\,d^2/4 = \sigma_L\,\pi\,d\,t \Rightarrow \sigma_L = p\,d\,/\,4t$

$$\sigma_c = p\,d\,/\,2\,t \qquad \sigma_L = p\,d\,/\,4\,t = \sigma_c\,/\,2$$

Hoop stress is twice the longitudinal stress - so longitudinal joints must be stronger than circumferential joints.

Maximum shear stress (in the plane): $\tau_{max} = (\sigma_c - \sigma_L)/2 = p\,d\,/\,8t$. Including the radial stress it is $p\,d\,/\,4t$.

3. Efficiency of joints

If the shell has riveted/welded joints, the plate is weakened:

$\sigma_c = p\,d\,/\,(2 + \eta_l)$ (η_l = efficiency of longitudinal joint)

$\sigma_L = p\,d\,/\,(4 + \eta_c)$ (η_c = efficiency of circumferential joint)

4. Strains and volume change

Using σ_c and σ_l as biaxial stresses (Poisson effect):

$$\varepsilon_c = (\sigma_c - \mu\,\sigma_l)/E = (p\,d/4t)(2 - \mu)$$
$$\varepsilon_l = (\sigma_l - \mu\,\sigma_c)/E = (p\,d/4t)(1 - 2\mu)$$
$$\varepsilon_v = 2\,\varepsilon_c + \varepsilon_l = (p\,d/4t)(5 - 4\mu)$$

- Circumferential strain $\varepsilon_c = \delta d / d$ ⇒ change in diameter $\delta d = \varepsilon_c\, d$
- Longitudinal strain $\varepsilon_l = \delta L / L$ ⇒ change in length $\delta L = \varepsilon_l\, L$
- Volumetric strain $\varepsilon_v = \delta V / V \Rightarrow \delta V = \varepsilon_v\, V$ ($V = \pi d^2 L / 4$)

Example 1

A thin cylinder $d = 800$ mm, $t = 10$ mm, $L = 2$ m carries $p = 2$ MPa. $E = 2\times10^5$ MPa, $\mu = 0.3$. Find stresses, strains and changes in d and V.

$\sigma_c = 2 \times 800 / (2 \times 10) = 80$ MPa, $\sigma_l = 40$ MPa

$\varepsilon_c = (80 - 0.3\times40)/2\times10^5 = 3.4\times10^{-4}$; $\varepsilon_l = (40 - 24)/2\times10^5 = 8\times10^{-5}$

$\delta d = 3.4\times10^{-4} \times 800 = 0.272$ mm, $\delta L = 8\times10^{-5} \times 2000 = 0.16$ mm

$\varepsilon_v = 2(3.4\times10^{-4}) + 0.8\times10^{-4} = 7.6\times10^{-4}$

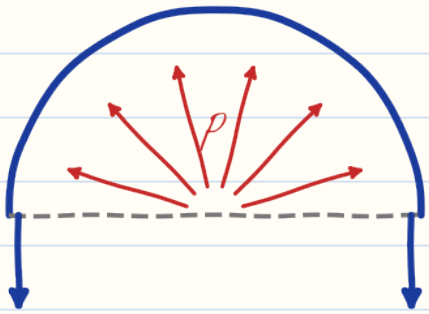
$V = \pi/4 \times 800^2 \times 2000 = 1.0053\times10^9$ mm^3 ⇒ $\delta V = 7.64\times10^5$ mm^3

Example 2: thickness

Find the thickness of a thin cylinder $d = 600$ mm for $p = 3$ MPa if $\sigma_{allow} = 100$ MPa.

$t = p\,d / 2\,\sigma = 3 \times 600 / (2 \times 100) = 9$ mm (use 10 mm)

5. Thin spherical shells



$\sigma = pd / 4t$

$p \pi d^2/4 = \sigma \pi d t \Rightarrow \sigma = p d / 4t$ (same in every direction)

$\varepsilon = (\sigma/E)(1 - \mu) = p d (1 - \mu) / 4 t E$ $\quad \varepsilon_v = 3 \varepsilon = 3 p d (1 - \mu) / 4 t E$

- $\delta d = \varepsilon d$, $\delta V = 3 \varepsilon V$ $(V = \pi d^3/6)$.
- For the same d, t and p the spherical shell has half the stress of the cylinder (hoop) - sphere is the most efficient pressure vessel shape.
- Cylinder with hemispherical ends: equal change of diameter requires $t_s/t_c = (1 - \mu)/(2 - \mu)$.

6. Wire-wound thin cylinders

A thin cylinder is wound with a layer of steel wire under initial tension. The wire puts the cylinder in initial circumferential compression, so that under working pressure the resultant stress is smaller - the cylinder resists higher pressure.

- Let wire stress σ_{wi} (tension) and cylinder stress σ_{ci} (compression) after winding, thicknesses t_w and t_c:
- Equilibrium after winding: $\sigma_{ci} t_c = \sigma_{wi} t_w$.
- Under pressure p the extra stresses are in the ratio of their moduli: $\Delta\sigma_w / E_w = \Delta\sigma_c / E_c$.
- Equilibrium of pressure: $\Delta\sigma_c t_c + \Delta\sigma_w t_w = p d / 2$.

$$\Delta\sigma_c = (p\,d/2)/(t_c + (E_w/E_c)\,t_w)$$

Final cylinder stress $= \Delta\sigma_c - \sigma_{ci}$ Final wire stress $= \sigma_{wi} + \Delta\sigma_w$

PART B - THICK CYLINDERS

7. Lamé's theory

When $t > d/20$ the hoop stress varies markedly across the wall, so it cannot be taken uniform. Lamé derived the stresses at radius r for a thick cylinder (internal radius r_1, external r_2).

- Assumptions: material homogeneous, isotropic, elastic; plane sections remain plane (longitudinal strain uniform); stresses are axisymmetric.
- Radial pressure p_r and hoop stress σ_θ at radius r:

$$p_r = B/r^2 - A \qquad \sigma_\theta = B/r^2 + A$$

(Lamé's equations)

A and B are constants found from the boundary conditions: $p_r = p_i$ at $r = r_1$, and $p_r = p_o$ at $r = r_2$:

$$A = (p_i r_1^2 - p_o r_2^2)/(r_2^2 - r_1^2) \qquad B = (p_i - p_o)\, r_1^2 r_2^2/(r_2^2 - r_1^2)$$

- Longitudinal stress (closed ends): $\sigma_L = A = (p_i r_1^2 - p_o r_2^2)/(r_2^2 - r_1^2)$.
- Radial stress is compressive and equals $-p_r$.
- Maximum shear stress $= (\sigma_\theta + p_r)/2$.

8. Internal pressure only ($p_o = 0$)

$$\sigma_\theta = (p_i r_1^2/(r_2^2 - r_1^2))(1 + r_2^2/r^2)$$
$$p_r = (p_i r_1^2/(r_2^2 - r_1^2))(r_2^2/r^2 - 1)$$

$\sigma_{\theta,max}$ (at r_1) $= p_i (r_2^2 + r_1^2)/(r_2^2 - r_1^2)$ σ_θ (at r_2) $= 2 p_i r_1^2/(r_2^2 - r_1^2)$

Example 3

Thick cylinder: internal dia 160 mm, external dia 240 mm, internal pressure 20 MPa. Find the stresses.

$r_1 = 80$, $r_2 = 120$. $A = 20 \times 6400 / (14400 - 6400) = 16$; $B = 16 \times 14400 = 230\,400$

σ_θ at inner $= 230400/6400 + 16 = 52$ MPa; at outer $= 230400/14400 + 16 = 32$ MPa

radial: at inner $= -20$ MPa (p_i); at outer $= 0$. $\sigma_L = A = 16$ MPa

τ_{max} (inner) $= (52 + 20)/2 = 36$ MPa

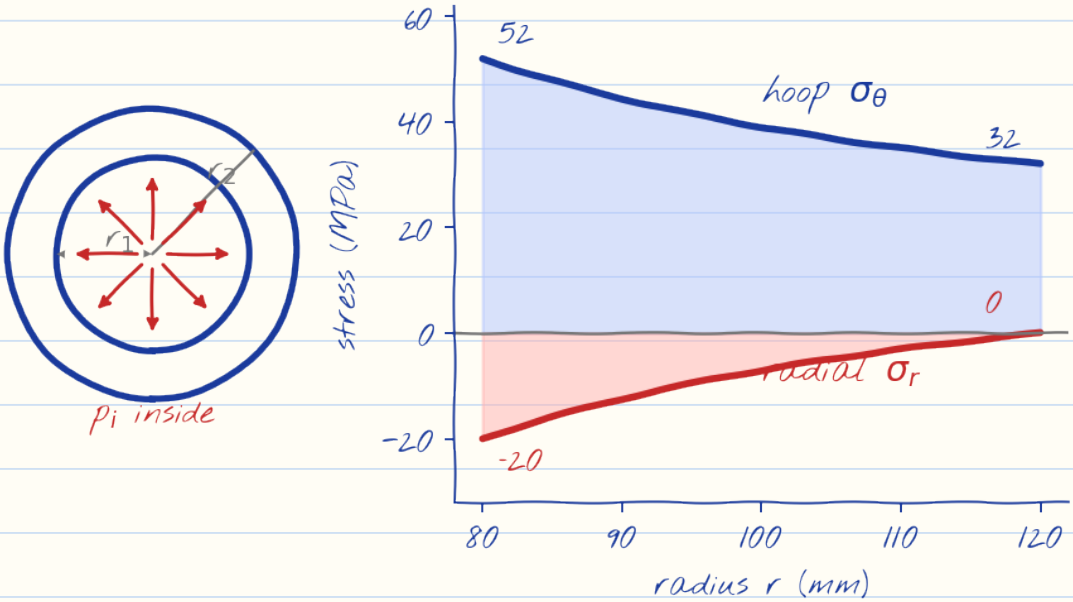


Fig: variation of hoop and radial stress across the wall (Example 3)

9. Design of thick cylinder

For a permissible hoop stress σ_t equate the maximum hoop stress at the inside:

$$r_2 = r_1 \sqrt{[(\sigma_t + p_i) / (\sigma_t - p_i)]} \qquad \text{thickness } t = r_2 - r_1$$

Note: if $p_i \geq \sigma_t$, no thickness is enough - thickening only helps up to a limit. Use a compound cylinder.

Example 4

$r_1 = 100$ mm, $p_i = 40$ MPa, $\sigma_t = 100$ MPa:

$r_2 = 100\sqrt{(140/60)} = 152.8$ mm $\Rightarrow$ $t = 52.8$ mm

10. External pressure and compound cylinders

- With both p_i and p_o use the general A and B above.
- Only external pressure p_o ($p_i = 0$): the maximum hoop stress is at the inner surface and is compressive: $\sigma_\theta = -2p_o r_2^2 / (r_2^2 - r_1^2)$.

Compound cylinder: an outer cylinder with a slightly smaller bore is heated and shrunk on to an inner cylinder (or pressed on). After cooling a shrink-fit pressure p_s acts at the junction.

- The inner cylinder is put in hoop compression, the outer in hoop tension.
- Under working pressure the stresses add to this initial state (superposition) - hoop stress across the wall becomes nearly uniform.
- The shrinkage allowance (difference of radii) = hoop strain difference at the junction x radius.
- Procedure: (i) hoop stress due to p_s in each cylinder with Lamé; (ii) stresses due to p_i on the combined cylinder; (iii) add algebraically.
- Benefit: lower peak stress, so the compound cylinder is stronger than a single one of the same material and thickness.

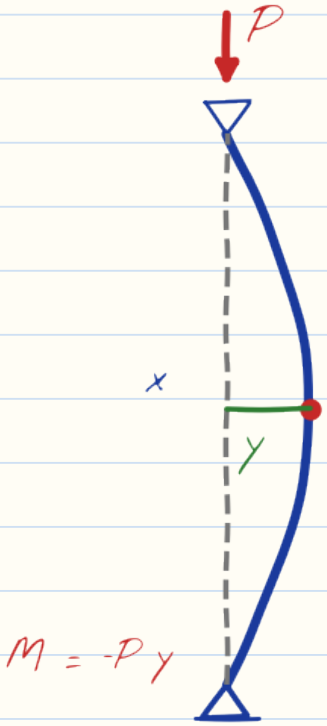
PART C - COLUMNS

11. Buckling and stability

A column (strut) is a compression member whose length is large compared with its lateral dimensions. A short column fails by crushing; a long slender column fails by buckling - sudden lateral bending at a load below the crushing load.

- Stable equilibrium: a small side push is recovered when removed.
- Unstable: the displacement keeps growing.
- Critical (buckling) load P_{cr}: the load at which the straight column is in neutral equilibrium – the axial load that can hold the column slightly bent.

12. Euler's formula – both ends pinned



Assumptions: column initially straight, axially loaded, material elastic and homogeneous, Euler's theory ignores the effect of axial shortening, and the failure is by buckling alone.

Bending moment at a section at distance x: $M = -Py$

$EI\, d^2y/dx^2 = -Py \Rightarrow d^2y/dx^2 + (P/EI)\, y = 0$

Solution: $y = C_1 \cos(\alpha x) + C_2 \sin(\alpha x)$, $\alpha^2 = P/EI$

$y = 0$ at $x = 0 \Rightarrow C_1 = 0$; $y = 0$ at $x = L \Rightarrow C_2 \sin(\alpha L) = 0 \Rightarrow \alpha L = n\pi$

Smallest load: $n = 1 \Rightarrow \alpha = \pi/L$

$$P_{cr} = \pi^2 E I / L^2$$

Buckling occurs about the axis of least moment of inertia, so use I_{min}.

13. Other end conditions – effective length

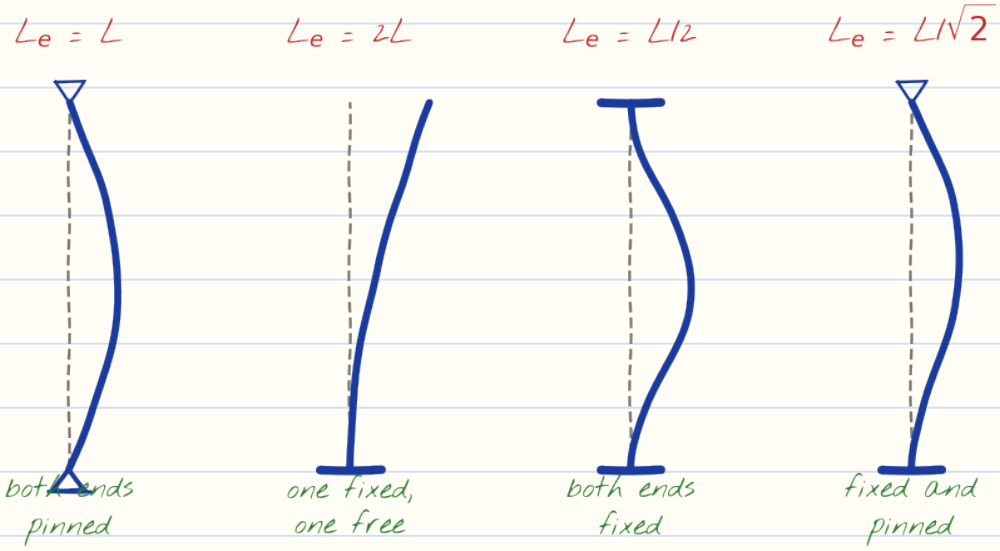


Fig: buckled shapes and effective lengths

$$P_{cr} = \pi^2 E I / L_e^2$$

End conditions	L_e	P_{cr}	Ratio
Both ends pinned	L	$\pi^2 EI/L^2$	1
One fixed, one free	$2L$	$\pi^2 EI/4L^2$	1/4
Both fixed	$L/2$	$4\pi^2 EI/L^2$	4
One fixed, one pinned	$L/\sqrt{2}$	$2\pi^2 EI/L^2$	2

14. Slenderness ratio and limitations of Euler's formula

Slenderness ratio $\lambda = L_e / r_{min}$ ($r = \sqrt{(I/A)}$ radius of gyration)

$$\sigma_{cr} = P_{cr}/A = \pi^2 E / \lambda^2$$

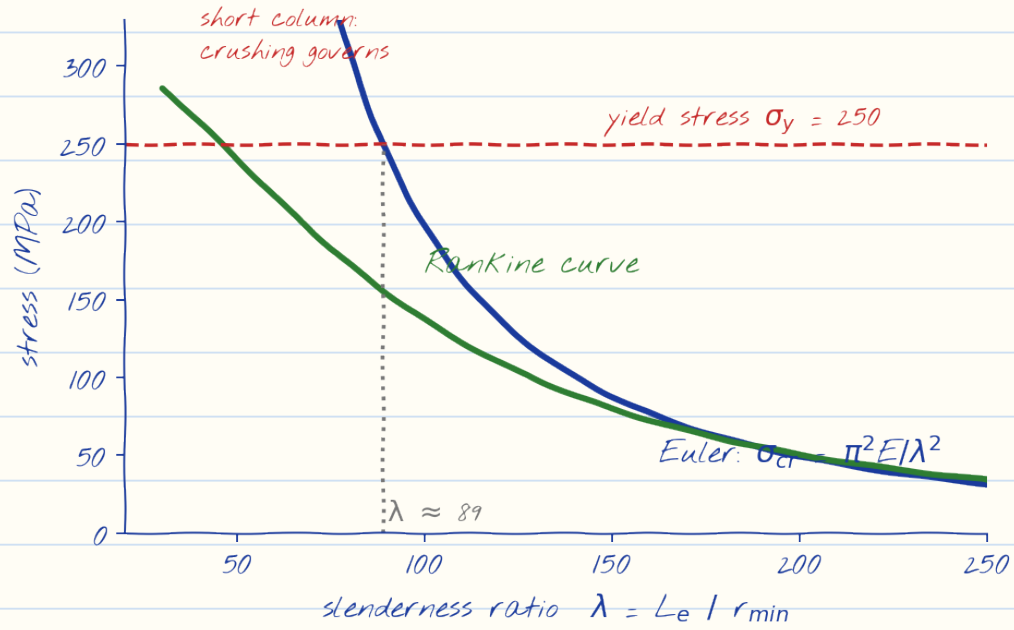


Fig: critical stress vs slenderness ratio (mild steel)

- Euler's formula is valid only while $\sigma_{cr} \leq$ the proportional limit (or yield stress): $\lambda \geq \lambda_{limit} = \sqrt{(\pi^2 E / \sigma_y)}$.
- For mild steel ($E = 2 \times 10^5$, $\sigma_y = 250$ MPa): $\lambda_{limit} \approx 89$ (often taken ≈ 80 to 100).
- For short columns (λ small) Euler gives a stress far above the yield stress - absurd; use Rankine or Johnson/Secant.
- The formula neglects initial crookedness, eccentricity of load and imperfect end conditions - so a factor of safety (about 3 for steel) is applied.

15. Rankine's formula

An empirical formula covering short as well as long columns. If $P_c = \sigma_c A$ is the crushing load and P_E the Euler load, then (Rankine-Gordon):

$$1/P_R = 1/P_c + 1/P_E \quad \Rightarrow \quad P_R = \sigma_c A / [1 + a (L_e/r)^2]$$

$$a = \sigma_c / (\pi^2 E) \text{ (Rankine constant)}$$

Material	σ_c (MPa)	a
Mild steel	320	1/7500
Cast iron	550	1/1600
Wrought iron	250	1/9000
Timber	50	1/750

- For small λ, $P_R \to \sigma_c A$ (crushing); for large λ, $P_R \to P_E$ (Euler).
- Rankine's formula is simple and widely used in design; safe load = P_R / FOS.

Example 5

A 3 m long solid steel column, 60 mm dia, both ends pinned. E = 200 GPa. Find P_{cr} (Euler) and the Rankine load (σ_c = 320 MPa, a = 1/7500).

$I = \pi \times 60^4 / 64 = 6.362 \times 10^5$ mm^4; $A = 2827.4$ mm^2; $r = d/4 = 15$ mm; $\lambda = 3000/15 = 200$

$P_{cr} = \pi^2 E I / L^2 = 9.8696 \times 2\times10^5 \times 6.362\times10^5 / 3000^2 = 139.5$ KN

$\sigma_{cr} = \pi^2 E/\lambda^2 = 49.3$ MPa ($< \sigma_y$ - Euler valid since $\lambda = 200 > 89$)

$P_R = 320 \times 2827.4 / (1 + 200^2/7500) = 904\,768 / 6.333 = 142.9$ KN

Example 6: end conditions

The same column with both ends fixed: $L_e = 1.5$ m, $\lambda = 100$.

$P_{cr} = 4 \times 139.5 = 558$ KN, $\sigma_{cr} = 197.4$ MPa (< 250, Euler still valid since $\lambda = 100 > 89$)

Example 7: hollow column

A hollow circular column D = 100 mm, d = 80 mm, L = 4 m, hinged at both ends, E = 200 GPa.

$I = \pi (100^4 - 80^4)/64 = 2.898\times10^6$ mm^4

$P_{cr} = \pi^2 \times 2\times10^5 \times 2.898\times10^6 / 4000^2 = 357.5$ KN

Safe load with FOS 3 = 357.5 / 3 = 119 KN

Quick revision - Unit V

$\sigma_c = pd/2t$ $\sigma_L = pd/4t$ sphere $\sigma = pd/4t$ $\varepsilon_v = (pd/4tE)(5 - 4\mu)$

Lame: $\sigma_\theta = B/r^2 + A$ $p_r = B/r^2 - A$ $P_{cr} = \pi^2 EI/L_e^2$ $P_R = \sigma_c A/(1 + a\lambda^2)$

- Important: derive σ_c and σ_L; change in volume of a thin cylinder; thin sphere.
- Important: Lame's equations and stress distribution; thickness of thick cylinder; compound cylinder.
- Important: Euler's formula for different end conditions; limitations; Rankine's formula problems.